

Predicting When Collection Statistics Help Top- k Selection: From Retrieval to Attention and Expert Routing

Team ANLP Doomsday

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Abstract

On sparse-autoencoder features, BM25 scores 0.582 nDCG@10 where a plain inner product over the same vectors scores 0.381. On SPLADE the ordering reverses, 0.455 against 0.568. What BM25 adds is collection frequency, the rate at which a feature fires across a corpus, and the published account of the reversal is that this helps only while a vocabulary stays Zipfian. That account has been observed one vocabulary at a time and never made predictive, so for a new sparse representation the only way to pick a scorer is to run both. We make the choice predictable. Eight concept vocabularies built from one frozen encoder are scored with a BM25-style function whose four components toggle independently, and we regress the BM25-minus-inner-product margin on six distributional measurements that cost one forward pass. Block-sparse attention and mixture-of-experts routing pose the same choice, since both score a query against learned non-negative features by inner product and neither consults selection frequency, so we apply the fitted predictor to them without refitting. Whether it holds there is an open question. Attention sinks fire for almost every query yet cannot be removed, and the rarest experts are the most critical, so frequency may signal importance differently inside a model than across a corpus. Evaluation runs on MS MARCO, LIMIT and BEIR, and on Hindi, Bengali, Telugu and Hindi-English code-mixed queries, where the distributions the correction depends on are known to shift.

1 Project Idea and Motivation

A learned sparse retriever scores a query-document pair as $\langle u_q, u_d \rangle$ over a concept vocabulary. A block-sparse attention kernel estimates how critical KV page i is for query q_t as $r_{t,i} = \langle q_t, c_i \rangle$ over page summaries, then attends to the top- k pages only (Tang et al., 2024). A mixture-of-experts router gates on $g_e = \langle h_t, w_e \rangle$ and sends the token to its

top- k experts (Shazeer et al., 2017; Fedus et al., 2022). These belong to three different literatures and perform one computation, an inner product against a learned non-negative basis followed by a hard top- k .

What all three leave out is any account of how often a given feature wins. Classical retrieval settled that question decades ago. The correction has three parts, namely inverse document frequency, which discounts features that fire everywhere, term-frequency saturation, so repeated evidence stops accumulating linearly, and length normalisation, so that large documents are not rewarded merely for being large (Robertson and Zaragoza, 2009). Parts of this correction have been reinvented inside the transformer under other names. Loss-free load balancing adds a per-expert bias driven by recent load (Wang et al., 2024), which is an additive, unnormalised frequency discount applied during training with no saturation or length term. The auxiliary losses it replaces push toward uniform expert use, which is a cruder version of the same idea (Zoph et al., 2022). Streaming inference hand-codes an exception for the highest-frequency keys in the cache (Xiao et al., 2024). Each of these is a local fix. No shared account exists of when a collection-level correction should help, and there is no way to tell before running the experiment.

Our project turns that into something measurable. We hold representation learning fixed, vary how a sparse basis is built and how it is scored, and ask whether cheap measurements of a basis predict how much the correction helps. The prediction is then tested in two places the measurements were never fitted on, both inside the transformer.

2 Background

BM25 scores a document d against query Q as

$$\sum_{i \in Q} \text{idf}_i \cdot \frac{f_i(d) (k_1 + 1)}{f_i(d) + k_1 \left(1 - b + b \frac{|d|}{\text{dl}}\right)}, \quad (1)$$

with $\text{idf}_i = \ln\left(1 + \frac{N - df_i + 0.5}{df_i + 0.5}\right)$. None of these ingredients requires words. Each needs only a representation with non-negative coordinates, a meaningful per-dimension collection frequency df_i , and a magnitude that admits a notion of length. Ordinary dense embeddings satisfy none of this. Coordinates are signed, every dimension is active in every document, and there is no length analogue, which is why dense retrieval is scored with a plain inner product instead.

Learned sparse features change the picture. Sparse-autoencoder activations, k -means codes and quantised codebooks are non-negative, sparse and count-like. The objects the transformer selects over are not themselves non-negative, since KV page summaries are built from signed keys, but the quantity IDF actually needs is a selection count, and that is non-negative by construction for both pages and experts. A page or an expert has a well-defined selection rate over a calibration set, and both admit a magnitude. All three settings are therefore eligible for the same correction, and whether it helps is an empirical question.

3 Related Work

Learned sparse retrieval and its vocabularies. SPLADE and its successors learn term weights over the backbone subword vocabulary and retain the inverted index (Formal et al., 2021; Lassance et al., 2024), with COIL taking a contextualised view of exact match (Gao et al., 2021) and a unified framework since organising the design space (Nguyen et al., 2023). A known pathology of this family is that some terms become badly over-activated, which degrades efficiency (Mackenzie et al., 2021). Recent work replaces the subword vocabulary with a learned concept vocabulary. CLSR indexes SAE-decomposed dense embeddings as concept units (Park et al., 2025), SPLARE trains an SAE-based sparse retriever directly (Formal et al., 2026), and SAE-SPLADE swaps SPLADE’s MLM head for an SAE encoder (Zong et al., 2026). All three score with an inner product.

Scoring learned sparse features with BM25. Clavié et al. (2026) show that SAE features taken from a frozen dense retriever have roughly Zipfian collection statistics and are directly BM25-scorable. Their Appendix C holds the sharpest evidence for our question. On the same suite, BM25 beats the inner product on SAE latent terms (0.582 against 0.381 nDCG@10) and loses to it on SPLADE-v3

(0.455 against 0.568), so the two scorers reverse order when the vocabulary changes beneath them. Han et al. (2026) run the comparison on image features and find (their §4.4) that the two differ by at most about two points of R@1, attributing the parity to top- k filtering already suppressing what IDF would down-weight. The same section reports that raising the per-patch activation budget from $k=16$ to $k=128$ makes document frequencies converge and drops R@1 from 0.508 to 0.038. Both papers that ran the comparison propose the same explanation, that BM25 helps while the vocabulary keeps its Zipfian discriminative structure. That hypothesis is therefore already in print and we are not supplying a missing explanation. It has only ever been observed on individual vocabularies, never tested across a controlled family of them, never quantified into anything usable before the fact, and never checked outside retrieval.

Selection inside the transformer. Quest scores KV pages by an inner product against page min-max key summaries and loads only the top- k (Tang et al., 2024). H2O keeps a mix of recent and heavy-hitter tokens (Zhang et al., 2023b), and StreamingLLM preserves the earliest keys as attention sinks (Xiao et al., 2024), which are entangled with the massive-activation mechanism (Sun et al., 2024). Expected Attention ranks a cached pair by how strongly future queries are predicted to attend to it and prunes the low scorers (Devoto et al., 2025). None of these uses the rate at which a page is selected across queries. On the routing side, Su et al. (2026) show that a handful of rare experts are disproportionately critical. To our knowledge no work treats retrieval scoring, KV selection and routing as instances of one scoring problem, and none tries to predict the effect of a correction before applying it.

4 Proposed Methodology

A modular scorer. For every setting we implement a single scoring function in which soft IDF weighting, term-frequency saturation, length normalisation and top- k sparsification are separate switches. The uncorrected inner product is the all-off configuration and BM25 is the all-on configuration, so we can ask which mechanism is responsible for any gain rather than only whether the bundle helps. The quantity we predict throughout is the margin, meaning the change in task quality between two configurations at matched budget.

Setting A: retrieval, used for calibration. We build eight concept bases from a single frozen encoder, spanning near-dense to strongly lexical, namely a split-rectified identity map, random projection, PCA, k -means codes, product quantisation (Jégou et al., 2011), residual quantisation, a Top- K sparse autoencoder (Gao et al., 2025), and SPLADE (Formal et al., 2021; Lassance et al., 2024). We include SPLADE on purpose, since the reversal we want to explain is SPLADE against SAE and omitting it would leave us unable to reproduce the phenomenon we study. Each basis is instantiated at pooled and token-level granularity, giving sixteen representations, so granularity becomes a variable rather than an unexamined confound.

Setting B: KV-page selection. We start from Quest-style page criticality (Tang et al., 2024) and add an online statistic sf_i , the fraction of prefill queries for which page i enters the top- k . The corrected score is

$$\tilde{r}_{t,i} = \phi(r_{t,i}) \cdot \text{idf}(\text{sf}_i) \cdot \ell(\|c_i\|), \quad (2)$$

where ϕ is the saturation term and ℓ the length analogue. The method is training-free and drops into an existing kernel. Whether attention sinks are exempted from the correction is an ablation rather than an assumption. If IDF suppresses them and accuracy falls, that outcome is the mechanism result.

Setting C: expert routing. The same correction is applied to router logits at inference time on OLMoE-1B-7B (Muennighoff et al., 2025), which activates 8 of 64 experts per layer and is open down to its training logs, with expert selection frequency measured on held-out calibration data. Loss-free balancing (Wang et al., 2024) is the additive, training-time special case we compare against. We check directly whether the correction protects or destroys the Super Experts identified by Su et al. (2026).

Diagnostics and the predictor. Before any selection experiment, and from a single forward pass over calibration data, we compute six measurements per basis, namely active fraction per item, the Zipf exponent of the selection-frequency curve, the Gini coefficient of activation mass, head mass, the variance of IDF values, and sink share, meaning the fraction of total score mass held by the top 1% most-frequently-selected features. Prior

work already reports several of these descriptively, with Han et al. (2026) fitting power-law exponents in $[1.20, 2.32]$ and head fractions under 2%, but none of it is used to predict anything. We regress the margin on these measurements using the many cheap points Setting A provides, validate by holding out whole basis families, and then apply the fitted model without refitting to Settings B and C. A predictor fitted on retrieval and tested inside a transformer is a stronger claim than one cross-validated within a single grid, and it is the part of the project a single-setting paper cannot produce.

Theory. We aim to characterise the conditions under which an IDF-weighted linear scorer over a learned basis provably reduces to a bilinear inner product on the underlying dense embeddings. We call this the collapse condition. Where it holds the two scorers must agree, which converts part of the grid into something explained rather than tabulated.

5 Datasets

Setting A fits its bases and tunes hyperparameters on MS MARCO passage (Bajaj et al., 2016), which supplies 8.8M passages, and also uses LIMIT (Weller et al., 2025), which separates lexical from semantic signal by construction and so should make any BM25 advantage easy to see. Configurations that look interesting transfer zero-shot to a subset of BEIR (Thakur et al., 2021) and to TREC-DL 19/20, with hyperparameters tuned only on MS MARCO so that transfer tests generalisation rather than additional fitting.

Because collection frequency is a property of a language and a corpus rather than of a model, we extend Setting A to the Hindi, Bengali and Telugu portions of MIRACL (Zhang et al., 2023a) and to Hindi-English code-mixed queries, where transliteration and mixing distort the document-frequency curve. Neither prior SAE-based sparse retriever touches an Indic language, and the multilingual evaluation in Zong et al. (2026) covers six MIRACL languages of which none is Indian. Corpus-level measures of mixing already predict downstream behaviour on such data (Kodali et al., 2022), and our diagnostics are measures of the same kind, so this is a direct test of whether the fitted model captures a property of the distribution or an artefact of English.

Setting B uses RULER (Hsieh et al., 2024), needle-in-a-haystack and LongBench on Llama-3.2-1B/3B-Instruct and Qwen2.5-7B-Instruct. Set-

ting C uses OLMoE-1B-7B with WikiText for perplexity and MMLU and GSM8K for downstream accuracy.

6 Evaluation Plan

For Setting A we report nDCG@10, MRR@10 and Recall@100/1000 alongside index size, query-document FLOPs and latency, since a scoring function that improves quality while destroying the efficiency benefit of sparsity would be a much less useful finding. As a pipeline check we reproduce the Appendix C reversal of Clavié et al. (2026) within a stated tolerance before trusting our own numbers. Setting B reports accuracy at matched token budget against Quest (Tang et al., 2024) and H2O (Zhang et al., 2023b), plus the KV budget needed for a fixed accuracy; Setting C reports perplexity, MMLU, GSM8K and routing entropy.

The headline result is the diagnostics-to-margin regression, reported as held-out R^2 under leave-one-basis-family-out validation, followed by the cross-setting transfer error when the model fitted on Setting A is applied to Settings B and C. Component ablations switch each BM25 mechanism on and off to identify which are necessary for which basis.

The grid produces several hundred pairwise comparisons, so we fix the analysis in advance. The primary comparison is all-on against all-off per representation; component ablations are secondary. Significance uses a paired bootstrap over queries with Holm–Bonferroni correction within each family of comparisons, and we report effect sizes alongside p -values rather than p -values alone.

Confirming that BM25 helps on Zipfian vocabularies would only repeat what the cited papers already predict, so we state in advance which outcomes would be informative. The diagnostics might predict a large gain for KV-page selection where the measured effect is negative, which would show contribution to the output and discrimination between queries coming apart inside the transformer. A basis family might have diagnostics saying BM25 should help where it measurably does not, falsifying the Zipfian account as stated. A frequency-corrected router might recover part of the auxiliary-loss benefit with no retraining.

7 Timeline and Division of Labour

Gupta builds the retrieval infrastructure, the indexing pipeline and the eight bases. Singh builds the

Week	Milestone
1	Frozen encoder; MS MARCO index built
2–3	Pooled grid; reproduce Appendix C
4–5	Token-level grid; MIRACL and code-mixed
6	Setting B built and run
7	Setting C routing (Mid, 30 Sep)
8	Fit predictor; held-out-family validation
9	Cross-setting transfer; collapse condition
10	Ablations, significance (Final, 31 Oct)

Table 1: Ten-week plan, 20 August to 31 October 2026.

modular scorer and the diagnostics, and fits the predictor. Sankar takes Setting B. Tripathy takes Setting C and the collapse condition, with Singh. Table 1 gives the schedule against the two remaining course deadlines.

The hardware available to us provides nodes of four GPUs with 11 GB of device memory each, 128 GB of host RAM and several terabytes of local scratch. That memory ceiling shapes the design. MS MARCO is encoded once with a frozen DistilBERT-scale encoder, a job that shards cleanly across nodes because passages are independent, and all eight pooled bases are then linear or quantisation maps over the cached embeddings, so no re-encoding is needed. Token-level granularity is capped at a 1M-passage subsample, roughly 55M vectors stored fp16 as a memmap on node-local scratch, which fits the per-node budget with room to spare. The MIRACL slices are far smaller than MS MARCO.

Settings B and C are inference-only and every model is sized to fit inside a single node. Llama-3.2-1B runs in fp16 on one card, while Llama-3.2-3B and Qwen2.5-7B run 4-bit across two, leaving headroom for a 32K KV cache. We chose OLMoE-1B-7B for Setting C over a larger checkpoint such as Qwen3-30B-A3B because its 6.9B total parameters fit a node at 4-bit where the 30B model would not, and the routing question we ask does not need the larger model, since OLMoE already exposes 64 experts per layer. If time runs short we cut in this order, token-level granularity for the quantisation bases first, then TREC-DL, then the longest RULER context lengths.

8 Risks and Limitations

If the predictor is weak on held-out families, the project becomes a negative result together with a resource. The diagnostics failing to predict the margin is informative precisely because two published papers assert the mechanism, and we would report

the grid as a benchmark with the collapse condition as the theoretical contribution. If Setting C proves infeasible on the hardware we can obtain we fall back to Settings A and B, which still support the transfer claim. Our core citations are months old, so being scooped is a real risk; the cross-setting transfer measurement is least exposed to it.

Four limitations are worth stating now. Setting A uses a single frozen encoder, so basis effects and encoder effects are not fully separable. Setting C is inference-time only, so we can make no claim about training dynamics. The diagnostics are correlative, and the cross-setting transfer test is the only part of the design with any causal flavour. Finally, the code-mixed evaluation depends on query sets that are smaller and noisier than MS MARCO, so those results will carry wider confidence intervals than the English ones.

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